

Point Processing

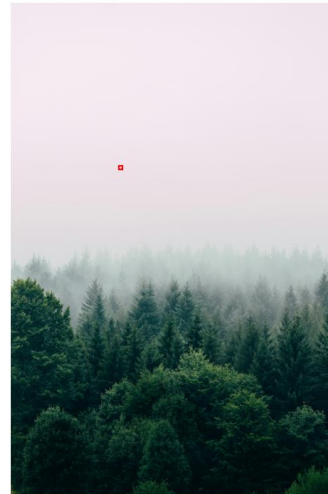
Histogram Equalisation and CLAHE

ARI 2129 | Principles of Computer Vision for AI

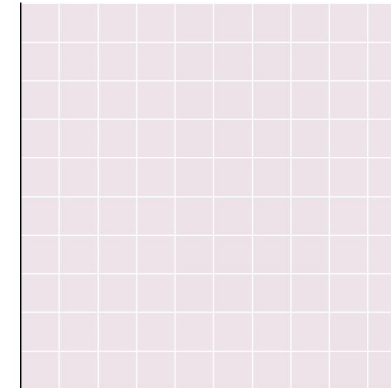
What is a Digital Image?

- A digital image is a grid of tiny squares, each one called a pixel
- Every pixel holds one number. Zero is black, 255 is white, anything else is grey
- A 1080p photo has over 2 million of these numbers
- Image processing is just maths on those numbers

Original Photo



Zoomed In — Individual Pixels



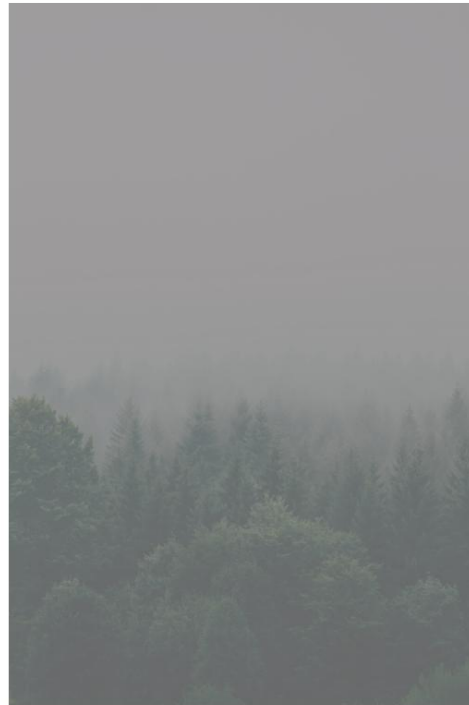
Each Pixel = One Number (0-255)

231	231	231	231	230
231	231	231	231	230
231	231	231	231	230
231	231	231	231	230
231	231	231	231	230

What is Contrast?

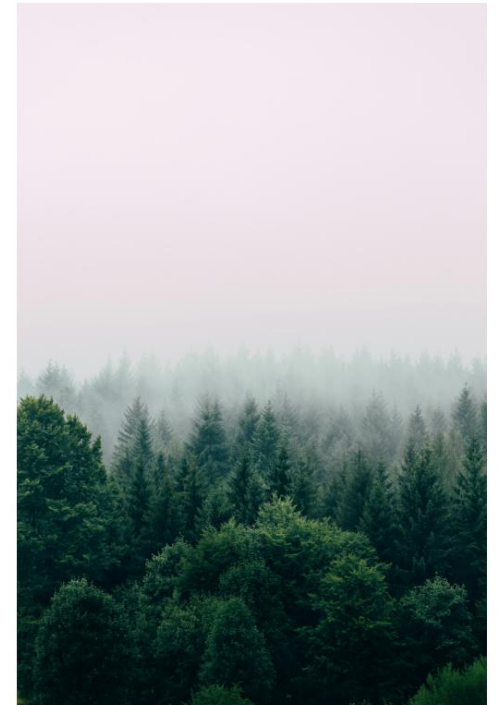
- Contrast is how spread out the pixel values are in an image
- High contrast: values run wide, the image looks sharp
- Low contrast: values are all bunched in a narrow range, everything looks flat
- Foggy photos are the obvious example. Medical scans have the same problem
- Without contrast, the eye cannot tell one thing from another

Low Contrast



Pixel values squeezed into a narrow range

Original (Better Contrast)

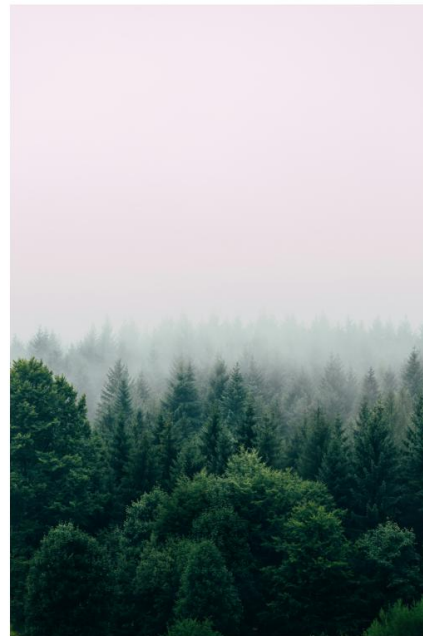


Pixel values spread across a wider range

What is a Histogram?

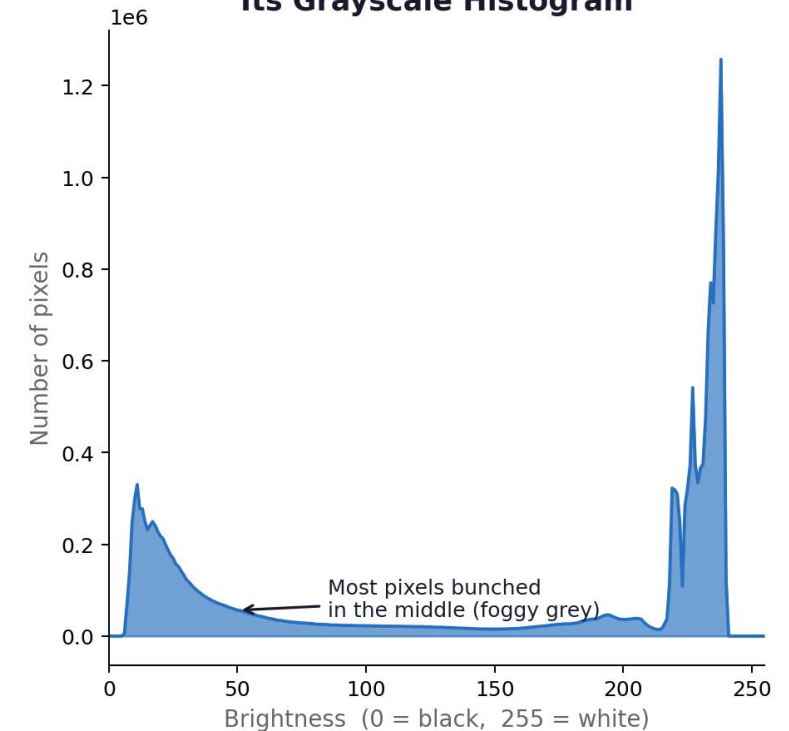
- A histogram counts how many pixels sit at each brightness level, 0 to 255
- Bars on the left: lots of dark pixels. Bars on the right: lots of bright ones
- A narrow spike means low contrast. A wide spread means good contrast
- You can read the problem off the chart before you even look at the photo
- Histogram equalisation reshapes this chart to cover the full range

Colour Photo (input)



Converted to grayscale for processing

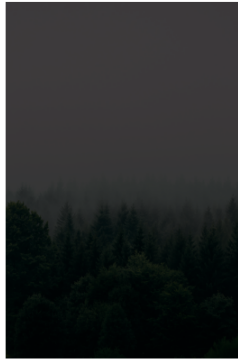
Its Grayscale Histogram



Reading a Histogram

Three shapes. Each one tells you something different about the image:

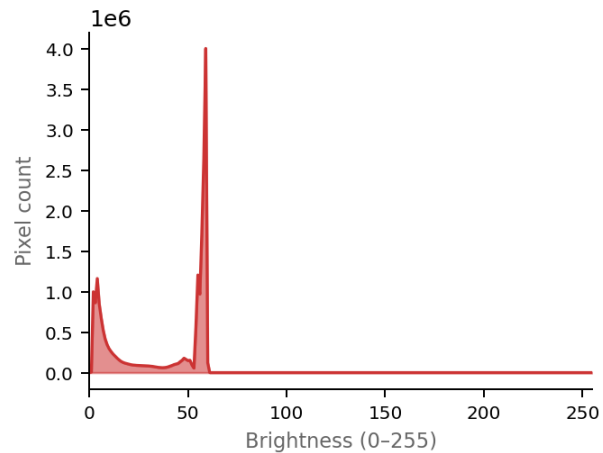
Dark / Underexposed



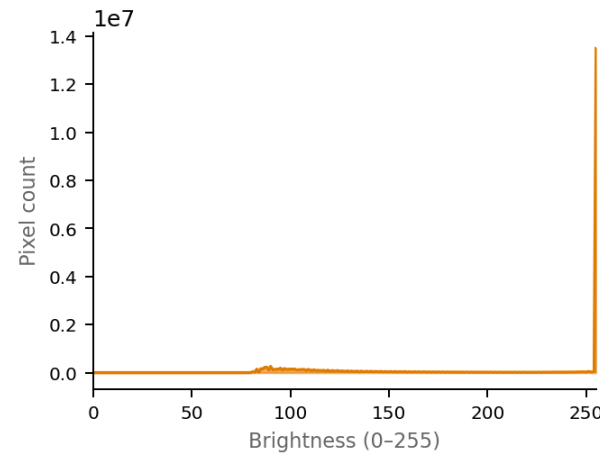
Overexposed



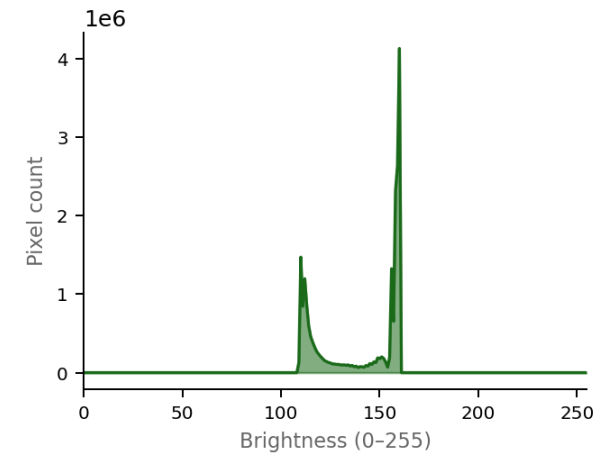
Low Contrast / Foggy



*Bars on the LEFT
(most pixels close to 0)*



*Bars on the RIGHT
(most pixels close to 255)*



*Narrow spike in MIDDLE
(all values similar)*

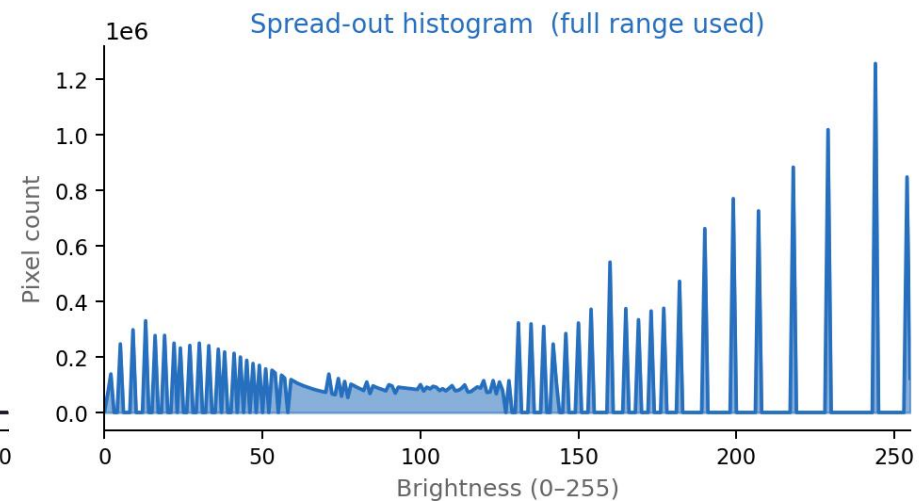
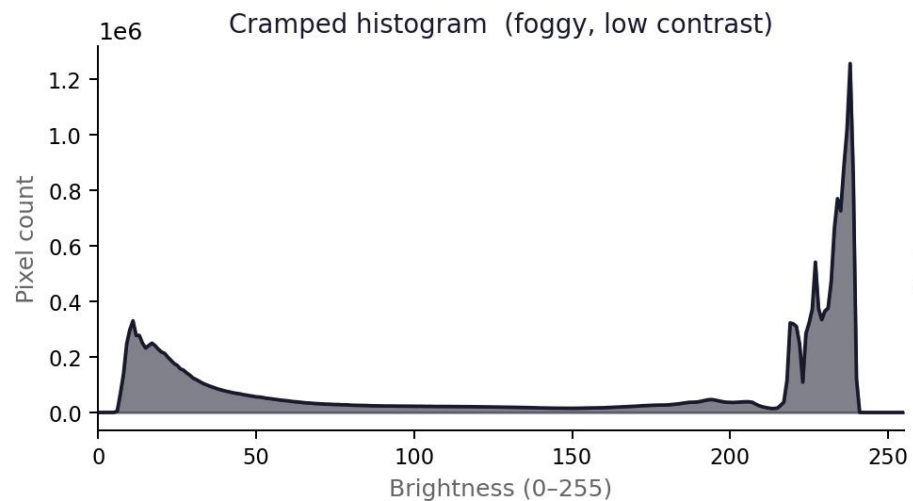
The Goal: Spread the Histogram

The algorithm reads the image's own histogram and remaps every pixel's brightness to cover the full 0 to 255 range. You give it an image, it works out the rest.

Before — Original



After — Histogram Equalised



Point Processing

What is Point Processing?

- Point processing is a basic image processing technique where each output pixel depends only on the corresponding input pixel.
- The same transformation rule is applied independently to every pixel in the image.
- No neighbouring pixels are involved.

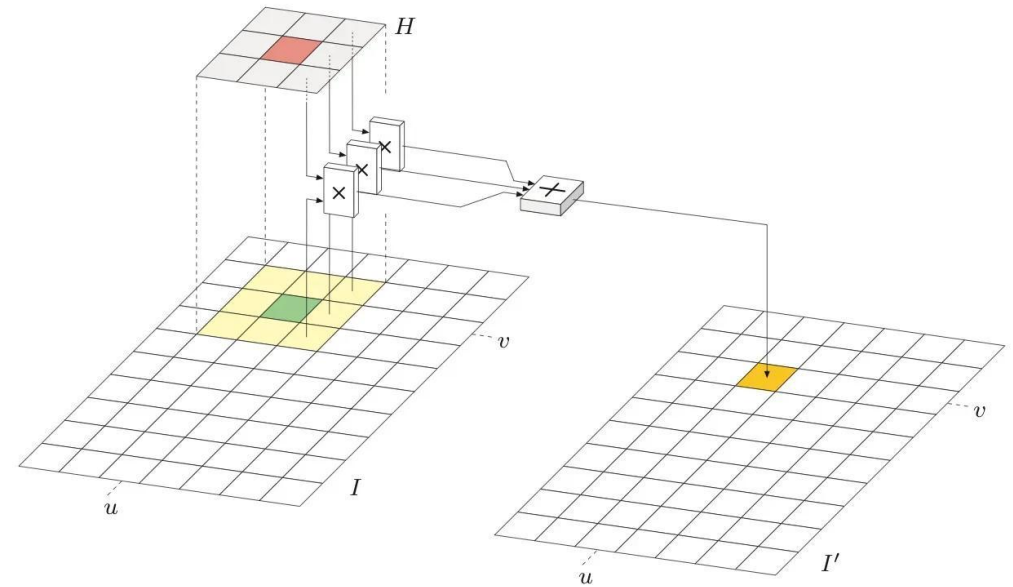
TRANSFORMATION FUNCTION

$$g(x, y) = T[f(x, y)]$$

● $f(x, y)$ = original pixel value
(input image)

● T = transformation function
(mapping rule)

● $g(x, y)$ = output pixel value
(output image)

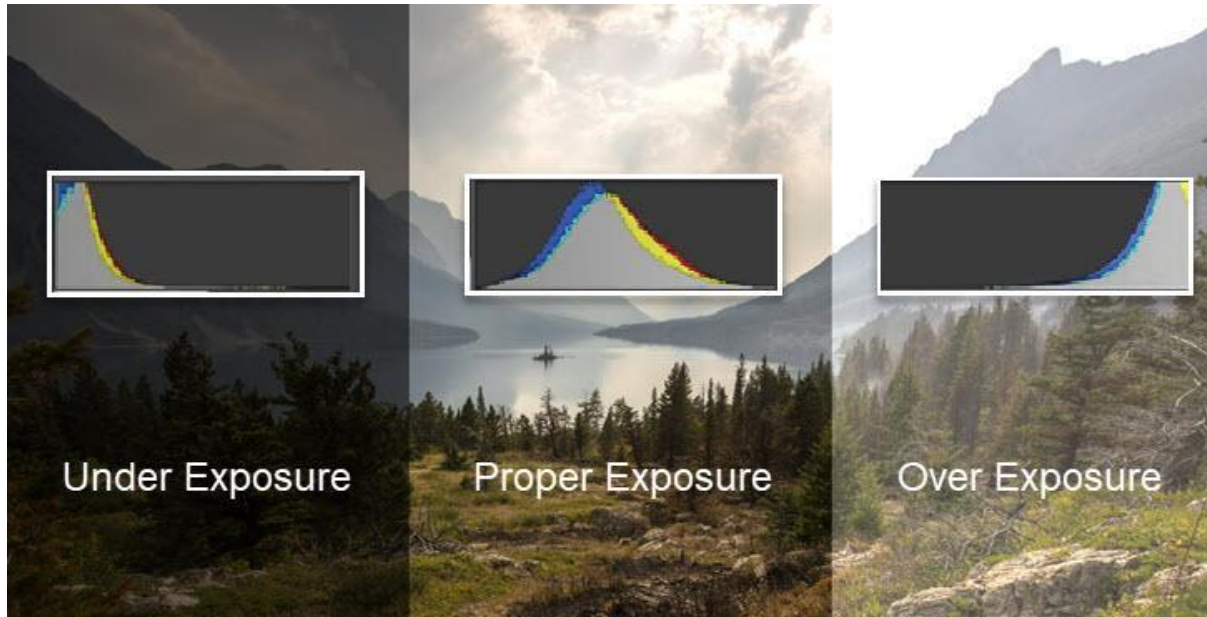


Key Idea: Histogram equalisation is still a point processing method because once the mapping function is created, every pixel is remapped independently.

The Normalised Histogram

What is a Histogram?

- A histogram counts how many pixels exist at each brightness level.
- In an 8-bit image there are 256 possible intensity levels from 0 to 255.



Key Idea: The normalised histogram tells us how image brightness values are distributed across the image.

The normalised histogram converts raw counts into probabilities:

$$p(r_k) = \frac{n_k}{N}$$

n_k = number of pixels with intensity r_k
 N = total number of pixels

The Cumulative Distribution Function (CDF)

What is the CDF?

$$CDF(r_k) = \sum_{j=0}^k p(r_j)$$

- The CDF is the running total of histogram probabilities.
- The CDF becomes the transformation function used in histogram equalisation.
- It spreads intensity values across a wider brightness range.

The CDF answers the question:

“How much of the image is this bright or darker?” A CDF just answers one question for every intensity level: "what fraction of all pixels are at or below this value"

Key Idea: Histogram equalisation works because the CDF redistributes pixel intensities more evenly across the available range.

Intensity	Count	Fraction	CDF
0	1	$1/4 = 0.25$	0.25
1	1	$1/4 = 0.25$	0.50
2	0	$0/4 = 0.00$	0.50
3	2	$2/4 = 0.50$	1.00

Deriving the Lookup Table

From CDF to New Pixel Values

Once the CDF is computed, it is scaled to the output brightness range:

$$s_k = \text{round}((L - 1) \times CDF(r_k))$$

For an 8-bit image:

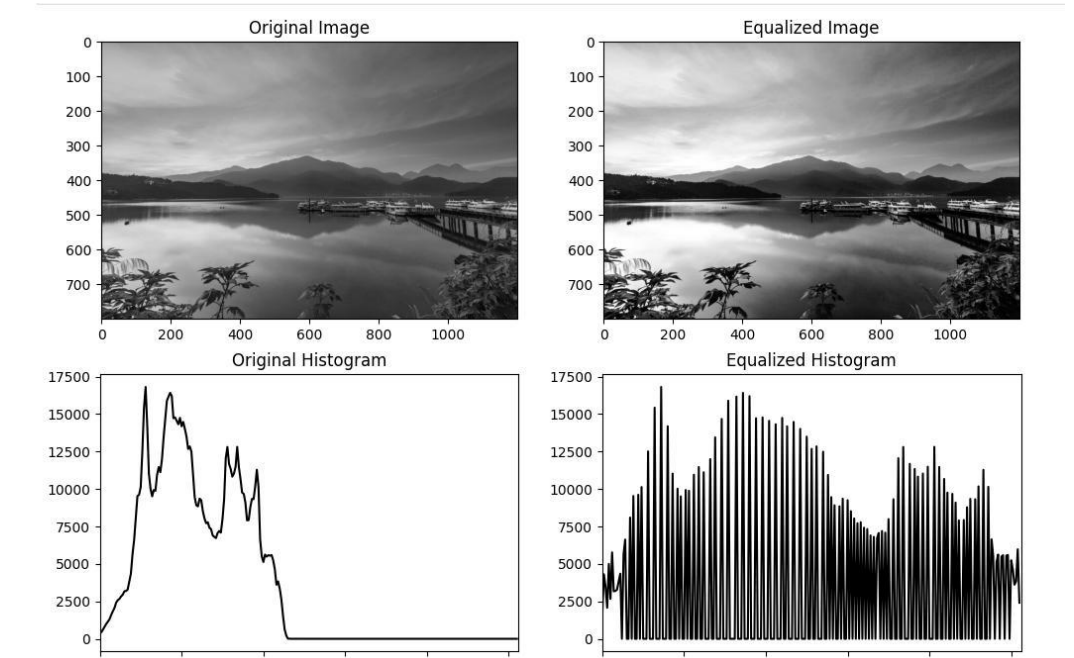
$$s_k = \text{round}(255 \times CDF(r_k))$$

What the Lookup Table Does

Maps old pixel values to new pixel values

Avoids recalculating the formula for every pixel

Makes image enhancement fast and efficient



Key Idea: The lookup table acts like a brightness translation table for the whole image.

Worked Numerical Example

Assume a small 4×4 image with only three brightness values:

10 appears 4 times , 50 appears 6 times, 80 appears 6 times, So total pixels=16

Step 1 — Normalise the Histogram

$$p(10) = \frac{4}{16} = 0.25$$

$$p(50) = \frac{6}{16} = 0.375$$

$$p(80) = \frac{6}{16} = 0.375$$

Step 2 — Compute the CDF

$$CDF(10) = 0.25$$

$$CDF(50) = 0.625$$

$$CDF(80) = 1.0$$

Step 3 — Build the Lookup Table

$$T(10) = 64$$

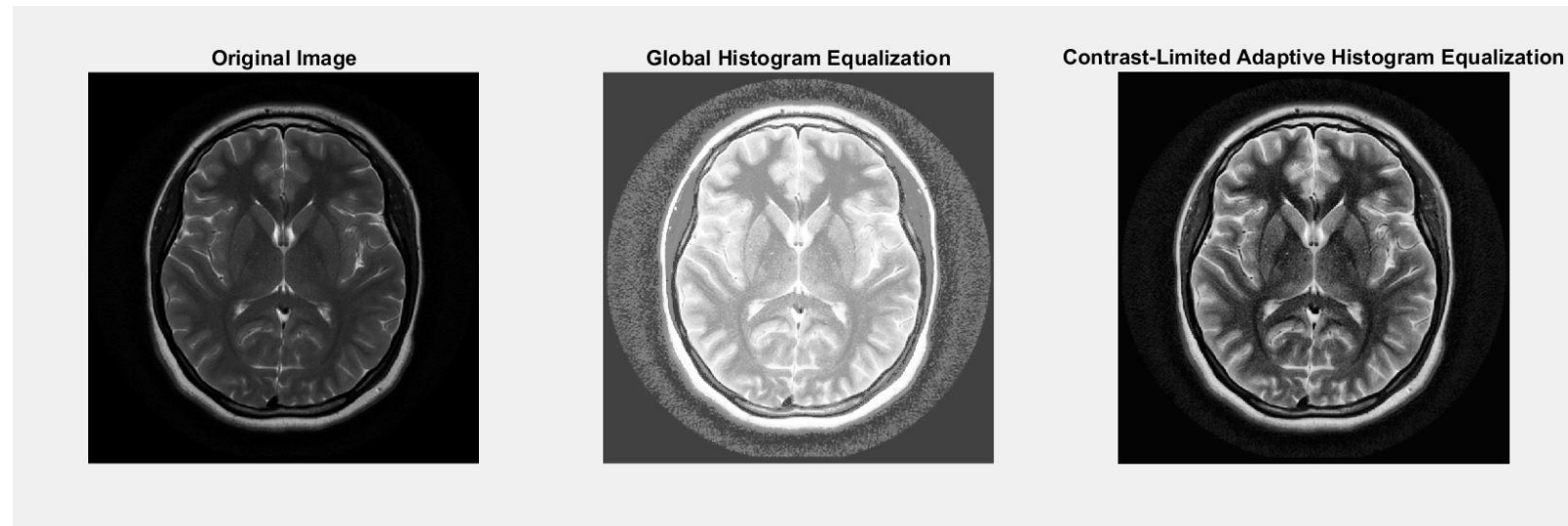
$$T(50) = 159$$

$$T(80) = 255$$

Final Result: The image now spans a much wider brightness range.

Contrast is improved because the intensity values are more spread out.

Key Takeaway: Histogram equalisation can be summarised in three steps:
Count, Accumulate, Scale

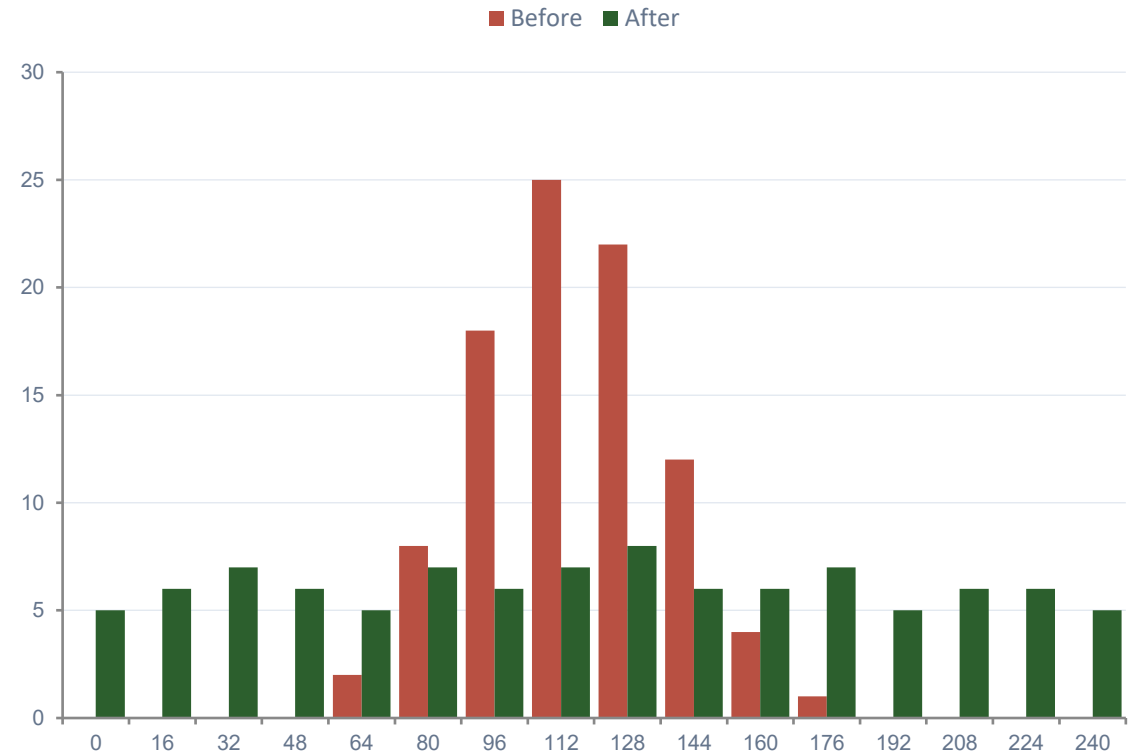


When HE Works

Best case: a uniformly dark or flat image

- Foggy scenes, underexposed portraits
- Pixels bunched in one zone
- No extreme spikes

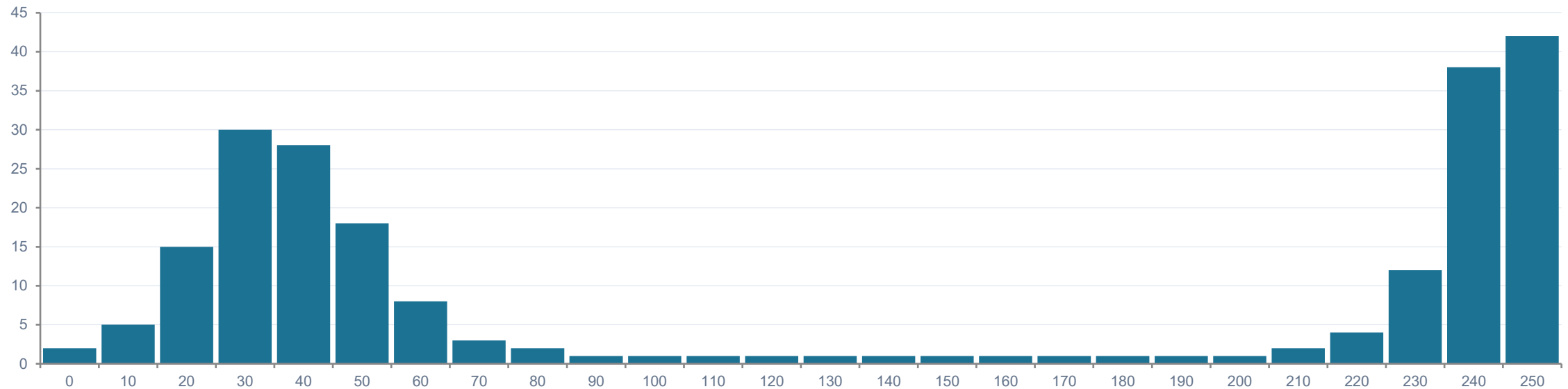
Histogram: before → after HE



For uniformly dull images, HE is very effective.

Failure Case: Bimodal Histograms

A concert photo — dark crowd around 30, bright LEDs near 250.



✓ Crowd becomes visible — dark cluster stretched.

✗ LEDs blow out to white — noise amplified.

HE treats the whole image as one. That's the problem.

Other Cases Where HE Breaks

Already high-contrast images — HE overshoots — clipping and artefacts.

Medical & scientific imaging — Subtle diagnostic gradients get crushed.

Large uniform regions — Sky or walls develop patchy banding.

Locally varying scenes — Global statistics don't fit local regions.

Equalising a good image breaks it.

When NOT to Use HE

Avoid HE when...

- the image is already well-exposed
- fine gradient detail matters (medical, scientific)
- large uniform regions dominate the frame
- lighting varies strongly across the image

Use instead...

- **CLAHE** — adaptive, tile-based
- **Gamma correction** — manual brightness curve
- **Histogram stretching** — gentler linear remap
- **Tone mapping** — for HDR

HE is a blunt tool. Sometimes you need a scalpel.

CLAHE: The Adaptive Fix

Contrast Limited Adaptive Histogram Equalisation

- 1 Divide into tiles**
Split the image into small non-overlapping blocks, typically 8×8.
- 2 Equalise each tile**
Apply histogram equalisation independently inside every tile.
- 3 Blend the seams**
Bilinear interpolation at the borders prevents blocky artefacts.

CLAHE thinks locally, not globally.

The Clip Limit

Problem

Small tiles have noisy histograms.

When we equalise them, noise gets amplified along with the real signal.

Result: grainy, over-enhanced patches — especially in smooth regions like sky or skin.

Solution: Clip Limit

Before computing the CDF, cap every histogram bin at a threshold β .

Any counts above β are collected as "excess" and redistributed evenly across all 256 bins.

This flattens the CDF slope $\rightarrow$ limits the maximum contrast amplification any pixel can receive.

OpenCV default: clipLimit = 2.0 (often called "clip factor" in papers).



Higher clip limit $\rightarrow$ stronger enhancement (more contrast, more noise risk). Lower $\rightarrow$ more conservative.

Formula: $\beta = (M/N)(1 + \alpha/100(s_{\max} - 1))$, α = clip factor 0–100%, M = pixels in tile, $N = 256$.

$\alpha = 0 \rightarrow$ identity (no change). $\alpha = 100 \rightarrow$ max slope $s_{\max}$ (typically 4 for X-ray).

"The clip limit controls how aggressive the enhancement is"

The tileSize Parameter

4 × 4

Small tiles

- Very local enhancement
- Can over-process fine detail
- More tiles → more histograms to compute
- Good for images with tiny structures

8 × 8

Default (OpenCV)

- 64 regions of ~64×64 px (for 512×512)
- Good balance: local enough to adapt, global enough to stay stable
- Recommended starting point
- Used in most CLAHE papers

16 × 16

Large tiles

- Closer to global HE behaviour
- Less local correction
- Fewer regions → faster
- Better for uniformly lit images

Both parameters interact: a small tile with a high clip limit will over-enhance; a large tile with a low clip limit barely changes anything. Always tune them together.

Bilinear interpolation is applied at tile borders to avoid visible seams between neighbouring regions.

"Tile size controls how local vs global the correction is"

CLAHE vs HE: Side by Side

Global HE Result

- ✗ Blown-out highlights (whites clip to 255)
- ✗ Amplified noise in uniform regions
- ✗ Global artefacts — whole image shifts

Uses one histogram for the entire image, so local detail is lost.

CLAHE Result

- ✓ Preserved highlights, no clipping
- ✓ Natural shadow recovery, no banding
- ✓ Noise controlled by clip limit

Each tile gets its own histogram → adapts to local contrast conditions.

	Global HE	CLAHE
Scope	Whole image at once	Tile by tile (e.g. 8×8 grid)
Noise	Amplifies noise everywhere	Clip limit caps amplification
Parameters	None	clipLimit, tileGridSize
Border artefacts	None (single region)	Removed via bilinear interpolation
Best for	Uniformly flat images	Complex / uneven lighting

"Same idea, smarter execution"

Real-World Applications



Medical Imaging

CLAHE reveals soft tissue detail in MRI/X-ray without destroying bone contrast. Used in CT-scans, mammography, retinal fundus images. Studies show improved Dice scores for spine segmentation when CLAHE is applied before U-Net.



Satellite Imagery

Enhances ground features in hazy or low-light aerial captures. Helps distinguish terrain, vegetation, and water bodies in remote-sensing images.



Night Vision / Surveillance

Lifts dark regions without blowing out street lights or headlights. Critical for CCTV analysis and UAV imagery under low-light conditions.



Face Detection Preprocessing

Normalises illumination across faces shot under mixed lighting. The Musa et al. study achieved 95% face recognition accuracy using CLAHE + LBPH at 98 lux.



Fingerprint Scanning

Enhances ridge detail for more reliable minutiae matching. Particularly useful for latent (partial) fingerprints captured in poor conditions.

"Anywhere lighting is uneven, CLAHE is useful"

Full Recap: The Teaching Journey

Complete Pipeline



Key Formulas

$$p(r) = h(r) / N$$

→ normalised histogram (probability of grey level r)

$$\text{CDF}(r) = \sum p(0..r)$$

→ cumulative distribution function (monotonically increasing)

$$T(r) = \text{round}(255 \times \text{CDF}(r))$$

→ the look-up table that maps old grey levels to new ones

✗ Failure: Global HE

Breaks bimodal histograms (e.g. dark object on bright background).

Over-enhances already well-exposed images.

Can cause washed-out, unnatural results.

✓ Fix: CLAHE

Local tiles → adapts to each region independently.

Clip limit → prevents noise amplification.

Bilinear interpolation → smooth transitions, no tile borders.

One-slide visual summary of the complete pipeline